



# Fuzzy Logic



# Fuzzy Learning

Fuzzy learning is a computational approach that utilizes fuzzy logic to handle uncertainty and imprecision in data.

This method is particularly useful in scenarios where traditional binary logic (true/false) is insufficient for making decisions based on vague or ambiguous information

# Fuzzy Logic

Fuzzy logic extends classical logic by allowing for degrees of truth rather than just binary true or false values.

It operates on the premise that many real-world situations cannot be adequately described using strict binary categories.

Introduced by Lotfi Zadeh in the 1960s, fuzzy logic provides a framework for reasoning that mimics human decision-making processes.

# Classical vs Fuzzy logic

Classical logic, also known as Boolean logic, is based on binary values: true or false, 1 or 0. It follows the principles of Aristotle's logic, where statements are either true or false with no middle ground.

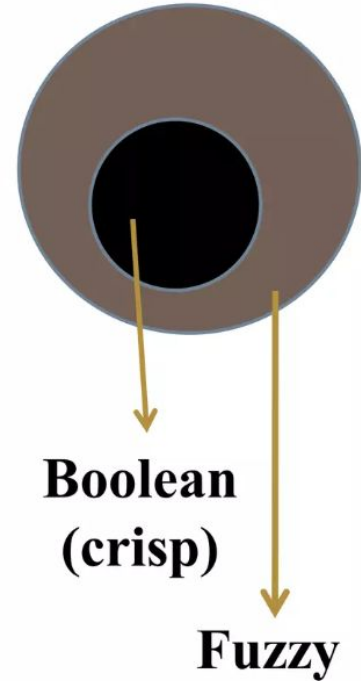
❑ Classical logic of Aristotle: Law of Bivalence “Every proposition is either True or False(no middle)”

❑ Jan Lukasiewicz proposed three-valued logic : True, False and Possible

❑ Finally Lofti Zadeh published his paper on fuzzy logic-a part of set theory that operated over the range  $[0.0-1.0]$

# Fuzzy Logic

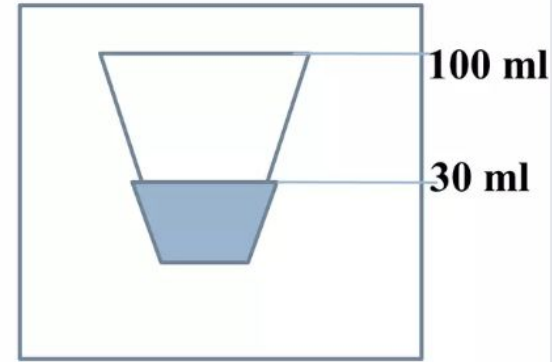
- ❑ Fuzzy logic is a superset of Boolean (conventional) logic that handles the concept of partial truth, which is truth values between "completely true" and "completely false".
- ❑ Fuzzy logic is **multivalued**. It deals with **degrees of membership** and **degrees of truth**.
- ❑ Fuzzy logic uses the continuum of logical values between 0 (completely false) and 1 (completely true).



For example, let a 100 ml glass contain 30 ml of water. Then we may consider two concepts: Empty and Full.

In boolean logic there are two options for answer i.e. either the glass is half full or glass is half empty.

In fuzzy concept one might define the glass as being 0.7 empty and 0.3 full.



# Fuzzy sets

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- It has an ability to classify elements into a continuous set using the concept of **degree of membership**.
- Fuzzy set is defined as a set whose elements have degrees of membership.
- The characteristics function or membership function not only gives 0 or 1 but can also give values between 0 and 1.
- Value 0  $\rightarrow$  non-membership
- Value 1  $\rightarrow$  complete membership
- Value between 0 and 1  $\rightarrow$  degree of membership

# Crisp and Fuzzy Set

- Classical set theory enumerates all element using

$$A = \{a_1, a_2, a_3, a_4, \dots, a_n\}$$

Set A can be represented by Characteristic function

$$\mu_a(x) = \begin{cases} 1 & \text{if element } x \text{ belongs to the set } A \\ 0 & \text{otherwise} \end{cases}$$

Example: Consider space X consisting of natural number  $\leq 12$

$$\text{Prime} = \{x \text{ contained in } X \mid x \text{ is prime number} = \{2, 3, 5, 7, 11\}$$

- In fuzzy set theory elements have varying degrees of membership

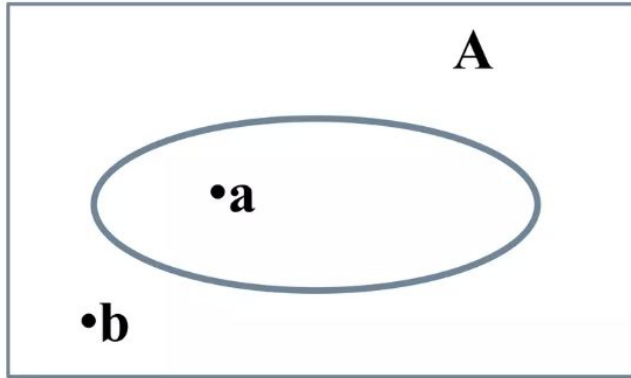
A fuzzy set can be represented by:

$$A = \{ \{x, A(x)\} \}$$

where,  $A(x)$  is the membership grade of a element  $x$  in fuzzy set

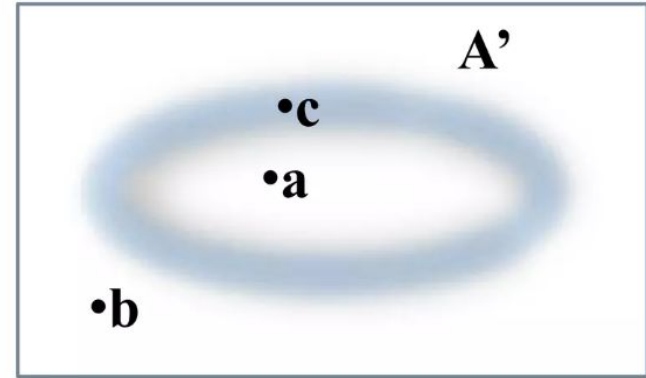
$$\text{SMALL} = \{ \{1, 1\}, \{2, 1\}, \{3, 0.9\}, \{4, 0.6\}, \{5, 0.4\}, \{6, 0.3\}, \{7, 0.2\}, \{8, 0.1\}, \{9, 0\}, \{10, 0\}, \{11, 0\}, \{12, 0\} \}$$

# Fuzzy and Crisp set



**Crisp set**

- a: member of crisp set  $A$
- b: not a member of set  $A$



**Fuzzy set**

- a: full member of fuzzy set  $A'$
- b: not a member of set  $A'$
- c: partial member of set  $A'$

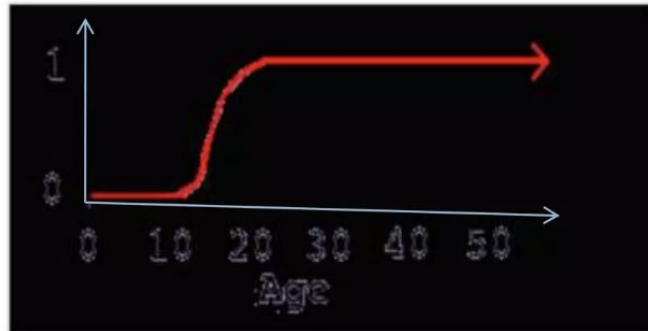
# Fuzzy and Crisp set

Crisp set

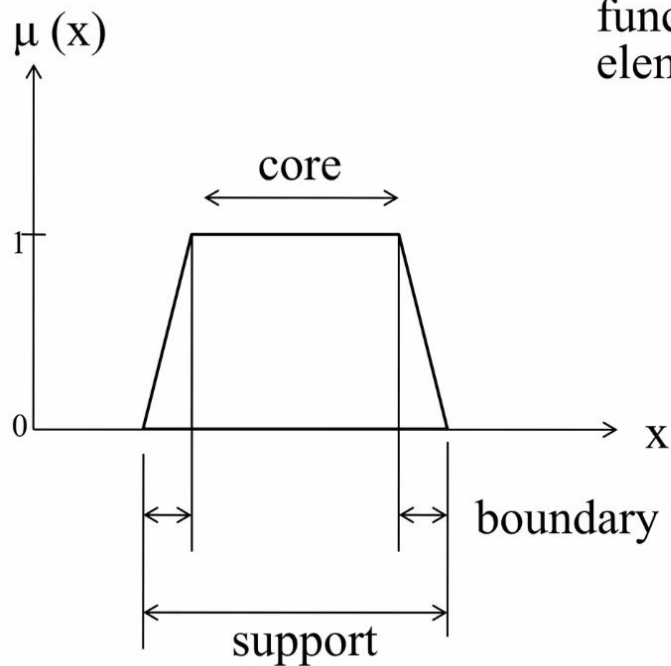
| Name     | Age | Degree of membership |
|----------|-----|----------------------|
| Sally    | 5   | 0                    |
| Jenny    | 18  | 0                    |
| Christen | 25  | 1                    |

Fuzzy set

| Name     | Age | Degree of membership |
|----------|-----|----------------------|
| Sally    | 5   | 0                    |
| Jenny    | 18  | 0.75                 |
| Christen | 25  | 1                    |



# Features of member function

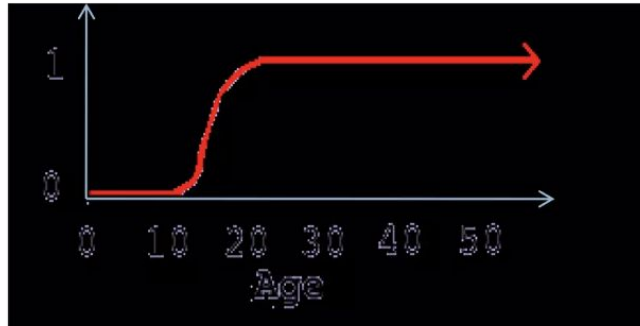


A membership function is a mathematical function which defines the degree of an element's membership in a fuzzy set.

- Core: region characterized by full membership in set  $A'$  i.e.  $\mu(x) = 1$ .
- Support: region characterized by nonzero membership in set  $A'$  i.e.  $\mu(x) > 0$ .
- Boundary: region characterized by partial membership in set  $A'$  i.e.  $0 < \mu(x) < 1$

# Membership Function

$$\text{adult}(x) = \begin{cases} 0, & \text{if age}(x) < 16\text{years} \\ (\text{age}(x) - 16\text{years})/4, & \text{if } 16\text{years} \leq \text{age}(x) \leq 20\text{years}, \\ 1, & \text{if age}(x) > 20\text{years} \end{cases}$$

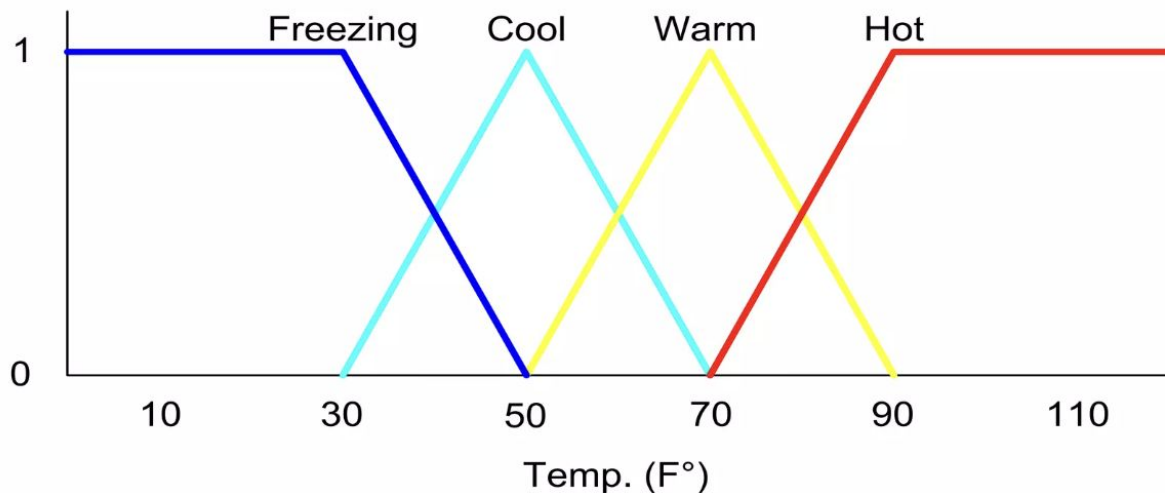


# Fuzzy Logic vs Probability

- ❑ Both operate over the same numeric range and at first glance both have similar values: 0.0 representing false (or non-membership) and 1.0 representing true.
- ❑ In terms of probability, the natural language statement would be "there is an 80% chance that Jane is old."
- ❑ While the fuzzy terminology corresponds to "Jane's degree of membership within the set of old people is 0.80."
- ❑ Fuzzy logic uses truth degrees as a mathematical model of the vagueness phenomenon while probability is a mathematical model of ignorance.

# Fuzzy Linguistic Variable

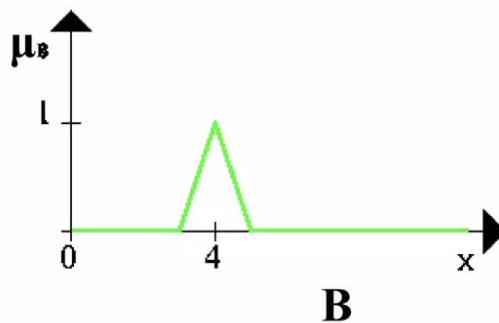
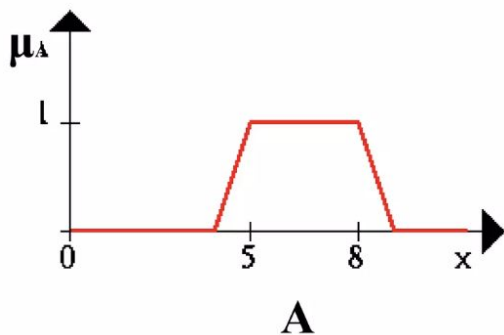
- ❑ Fuzzy Linguistic Variables are used to represent qualities spanning a particular spectrum
- ❑ Temp: {Freezing, Cool, Warm, Hot}



# Operations on Fuzzy set

Consider:

$$A = \{1/2 + .5/3 + .3/4 + .2/5\} \quad B = \{.5/2 + .7/3 + .2/4 + .4/5\}$$

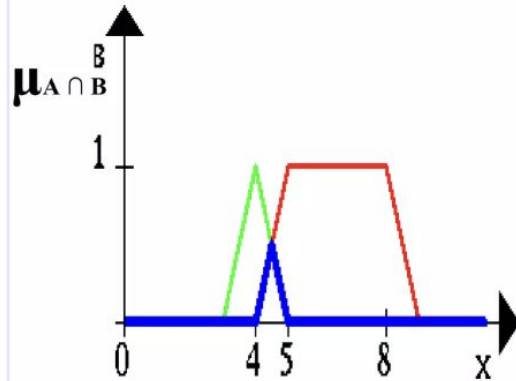


—> Fuzzy set (A)

—> Fuzzy set (B)

—> Resulting operation of fuzzy sets

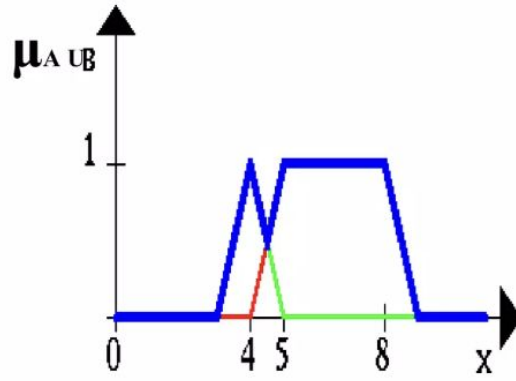
## INTERSECTION (A ^ B)



$$\mu_{A \cap B} = \min(\mu_A(x), \mu_B(x))$$

$$\{.5/2 + .5/3 + .2/4 + .2/5\}$$

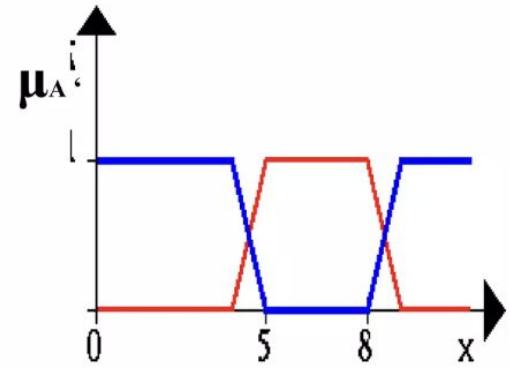
## UNION (A v B)



$$\mu_{A \cup B} = \max(\mu_A(x), \mu_B(x))$$

$$\{1/2 + .7/3 + .3/4 + .4/5\}$$

## COMPLEMENT (¬A)



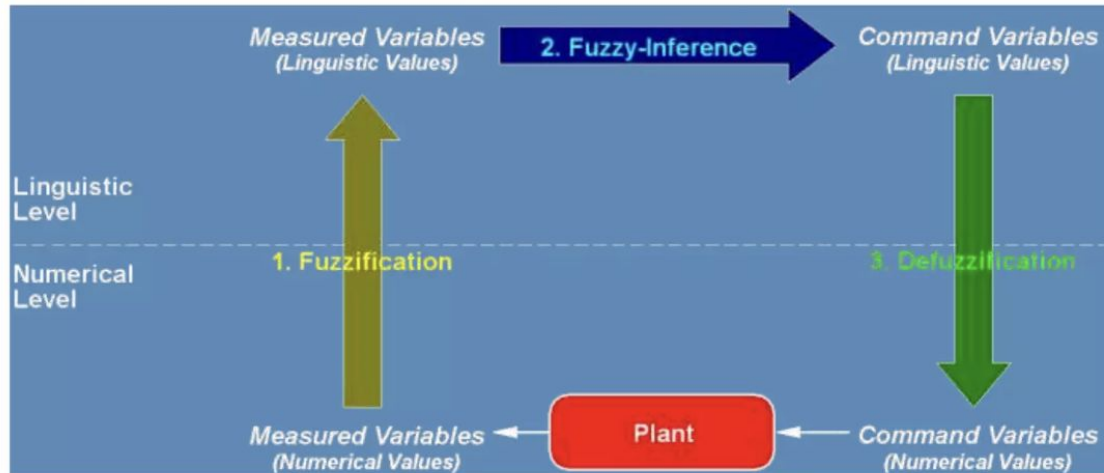
$$\mu_{A'} = 1 - \mu_A(x)$$

$$\{1/1 + 0/2 + .5/3 + .7/4 + .8/5\}$$

# Example: Speed Calculation

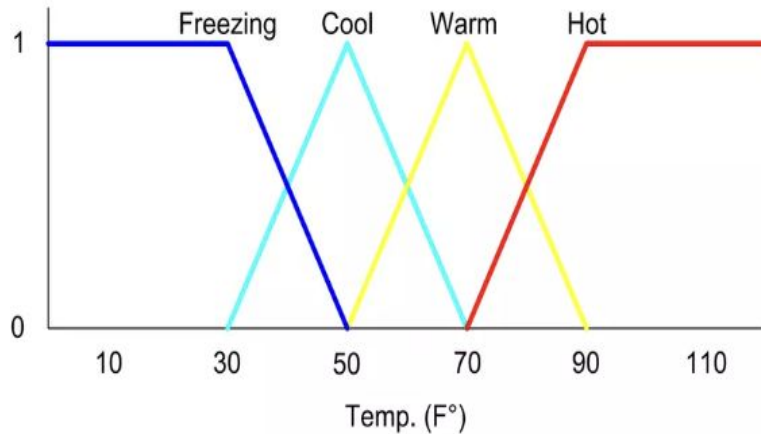
How fast will I go if it is

- ❑ 65 F
- ❑ 25 % Cloud Cover ?

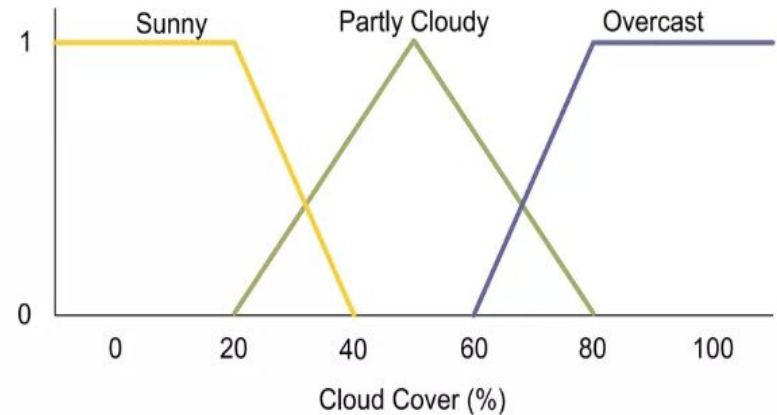


# Input:

**Temp: {Freezing, Cool, Warm, Hot}**

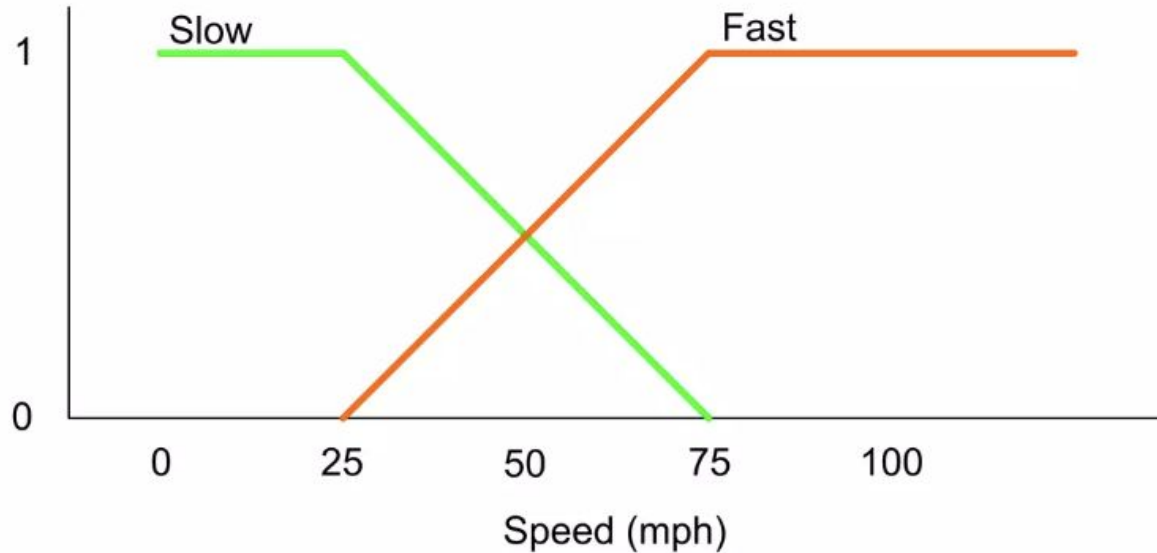


**Cover: {Sunny, Partly cloudy, Overcast}**



# Output

**Speed: {Slow, Fast}**

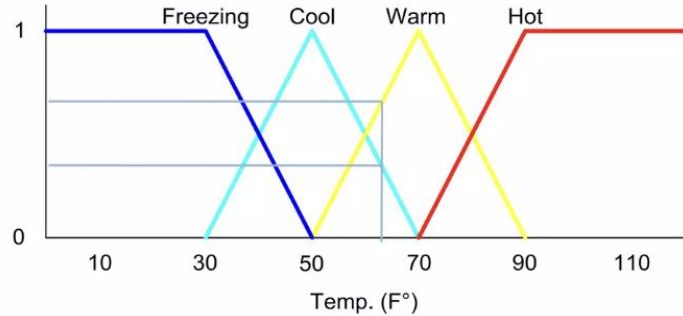


# Rules

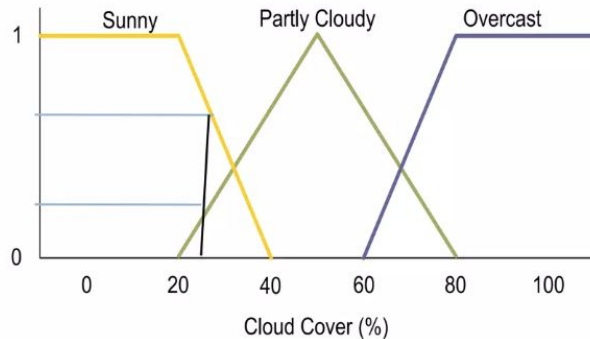
- ❑ If it's Sunny and Warm, drive Fast  
 $\text{Sunny}(\text{Cover}) \wedge \text{Warm}(\text{Temp}) \Rightarrow \text{Fast}(\text{Speed})$
- ❑ If it's Cloudy and Cool, drive Slow  
 $\text{Cloudy}(\text{Cover}) \wedge \text{Cool}(\text{Temp}) \Rightarrow \text{Slow}(\text{Speed})$
- ❑ Driving Speed is the combination of output of these rules...

# Fuzzification: Calculate Input Membership Levels

❑ **65 F  $\Rightarrow$  Cool = 0.4, Warm = 0.7**



❑ **25% Cover  $\Rightarrow$  Sunny = 0.8, Cloudy = 0.2**



# Calculation

- If it's Sunny and Warm, drive Fast

$\text{Sunny}(\text{Cover}) \wedge \text{Warm}(\text{Temp}) \Rightarrow \text{Fast}(\text{Speed})$

$$0.8 \wedge 0.7 = 0.7$$

$$\Rightarrow \text{Fast} = 0.7$$

- If it's Cloudy and Cool, drive Slow

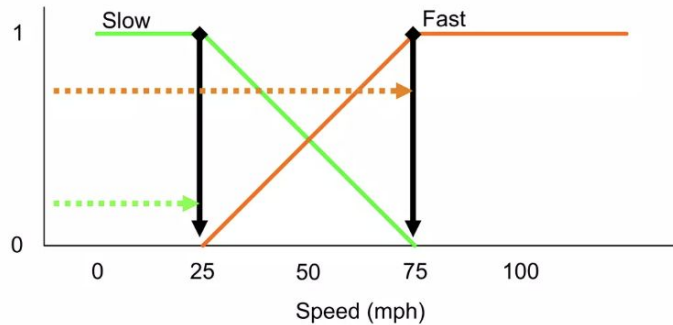
$\text{Cloudy}(\text{Cover}) \wedge \text{Cool}(\text{Temp}) \Rightarrow \text{Slow}(\text{Speed})$

$$0.2 \wedge 0.4 = 0.2$$

$$\Rightarrow \text{Slow} = 0.2$$

# Defuzzification: Constructing the output

- Speed is 20% Slow and 70% Fast



- Find centroids: Location where membership is 100%

- Speed = weighted mean  
=  $(2 \cdot 25 + 7 \cdot 75) / (9)$   
= 63.8 mph

# Fuzzy Application

- ❑ **Automobile and other vehicle subsystems** : used to control the speed of vehicles, in Anti Braking System.
- ❑ **Temperature controllers** : Air conditioners, Refrigerators
- ❑ **Cameras** : for auto-focus
- ❑ **Home appliances**: Rice cookers , Dishwashers , Washing machines and others

# Drawbacks

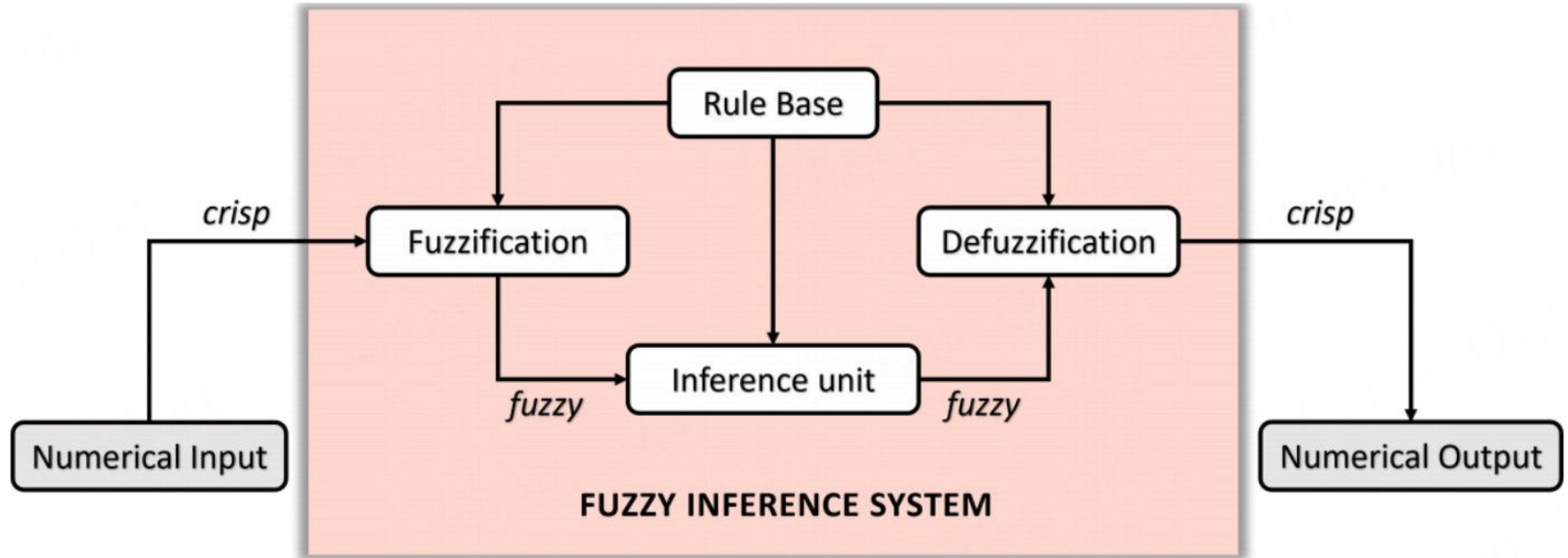
- ❑ Fuzzy logic is not always accurate. The results are perceived as a guess, so it may not be as widely trusted .
- ❑ Requires tuning of membership functions which is difficult to estimate.
- ❑ Fuzzy Logic control may not scale well to large or complex problems
- ❑ Fuzzy logic can be easily confused with probability theory, and the terms used interchangeably. While they are similar concepts, they do not say the same things.

# Fuzzy Inference System

A Fuzzy Inference System (FIS) is a framework that uses fuzzy logic to map inputs to outputs.

It processes input data in the form of fuzzy sets, applies rules, and produces a fuzzy output, which is often defuzzified into a crisp value.

# Structure of FIS



# Components of fuzzy inference system

**Fuzzification:** Converts crisp inputs into fuzzy sets by determining the degree to which they belong to various fuzzy sets.

**Knowledge Base:** Comprises a set of fuzzy rules and membership functions that define how inputs relate to outputs.

**Inference Engine:** Applies fuzzy logic rules (IF-THEN) to compute the fuzzy output by combining inputs.

**Defuzzification:** Converts the fuzzy output back into a crisp value for decision-making.

# Fuzzy Inference Method

Fuzzy inference methods are the ways in which inputs are processed using fuzzy rules to infer outputs in a Fuzzy Inference System (FIS).

These methods determine how fuzzy sets are combined to produce results.

# Fuzzy Inference Method: Mamdani Method

Most commonly used fuzzy inference method.

Employs min-max operations for inference.

Rules are expressed in the form of IF-THEN statements, and the final fuzzy output is defuzzified to get a crisp result.

Example: IF temperature is "high" AND humidity is "low," THEN fan speed is "medium."

